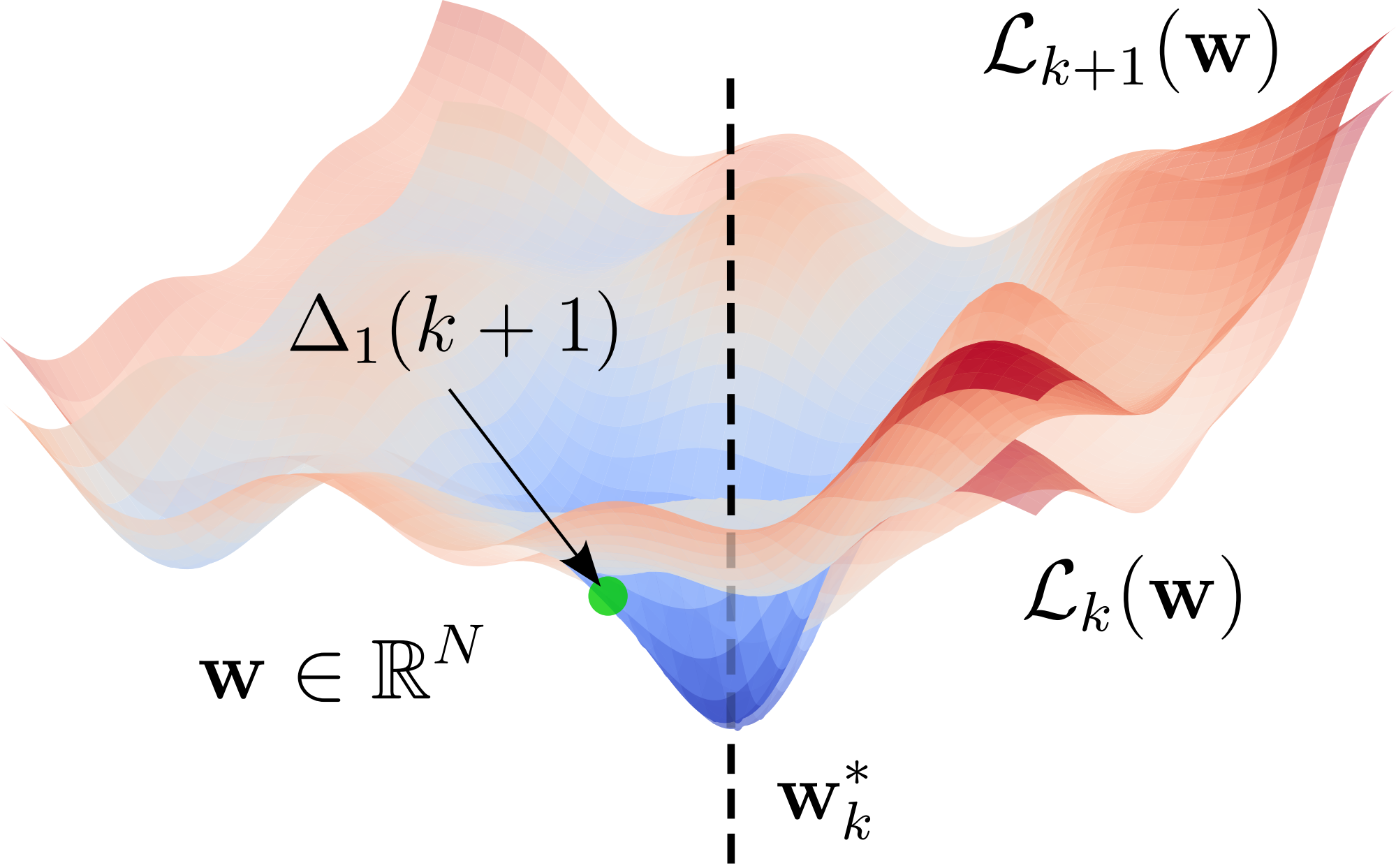
$\mathcal{L}_{k+1}(\mathbf{w})$
$\Delta_1(k+1)$
$\mathcal{L}_k(\mathbf{w})$
$\mathbf{w} \in \mathbb{R}^N$
$\mathbf{w}_k^*$